# *Course: Cryptography and Network Security*
# *Code: CSN15403*
# *Branch: B.Tech -5th Semester*

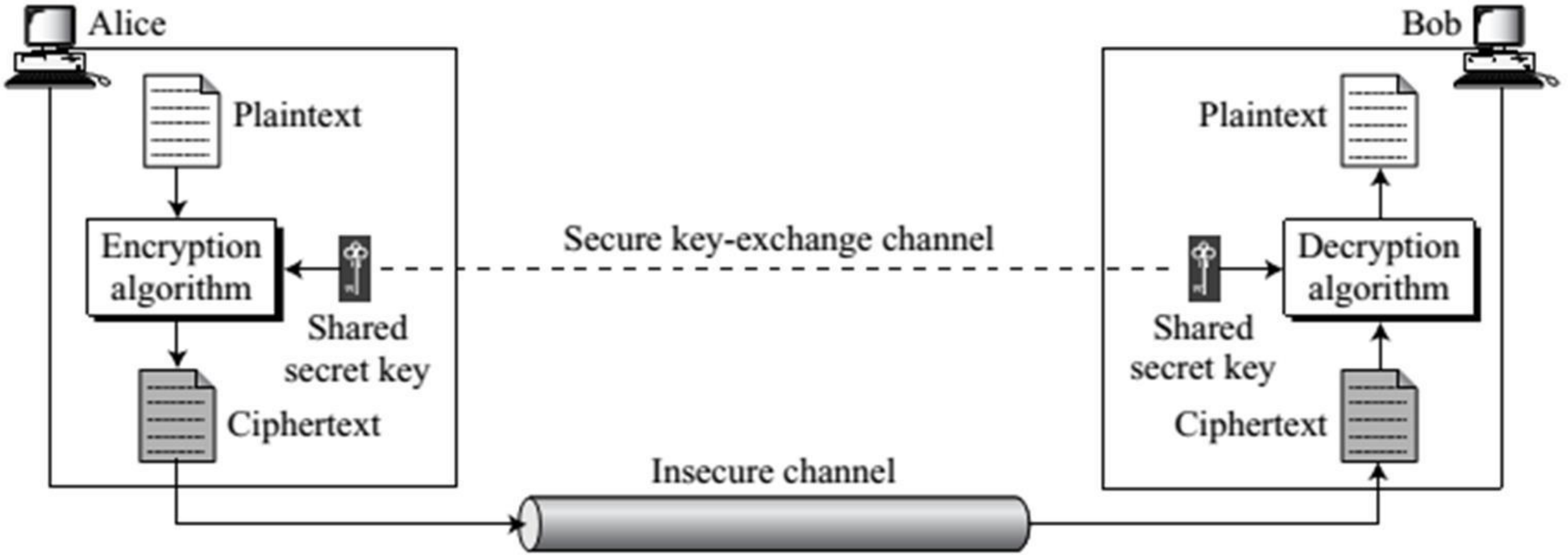
Lecture 5 – Traditional Symmetric-Key Ciphers

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# Symmetric-key Cipher



# Symmetric-key Cipher

- Symmetric-key encipherment uses a single key for both encryption and decryption.
- Encryption and decryption algorithms are inverses of each other.
- If P is the plaintext, C is the ciphertext, and K is the key, the encryption algorithm $E_k(x)$ creates the ciphertext from the plaintext; the decryption algorithm $D_k(x)$ creates the plaintext from the ciphertext.
- We assume that $E_k(x)$ and $D_k(x)$ are inverses of each other: they cancel the effect of each other if they are applied one after the other on the same input.

Encryption: $C = E_k(P)$ Decryption: $P = D_k(C)$

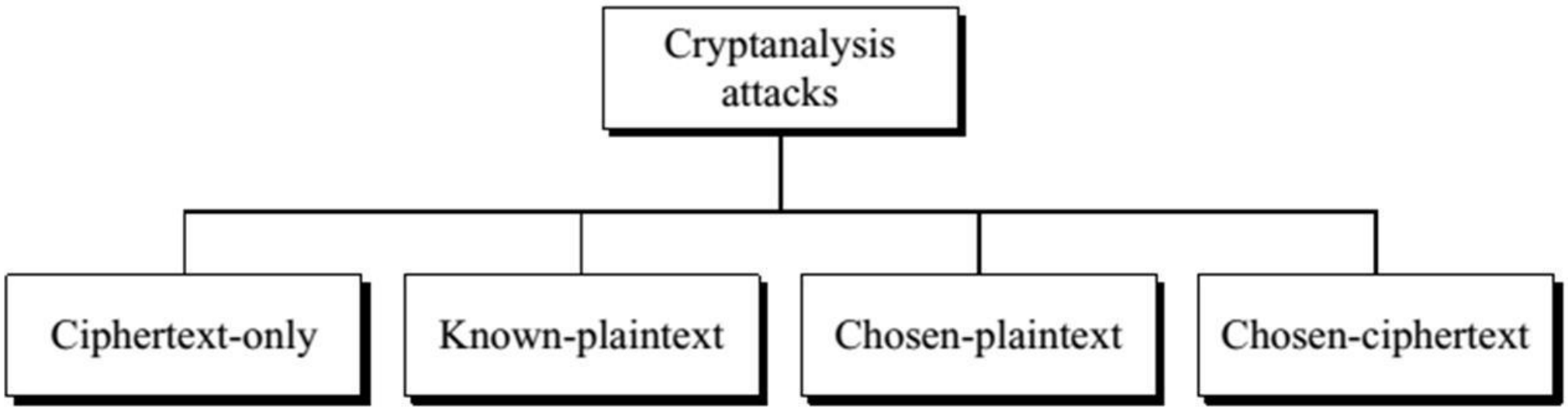
In which, $D_k(E_k(x)) = E_k(D_k(x)) = x$

**Alice:** $C = E_k(P)$ **Bob:** $P_1 = D_k(C) = D_k(E_k(P)) = P$

# Kerckhoff's Principle

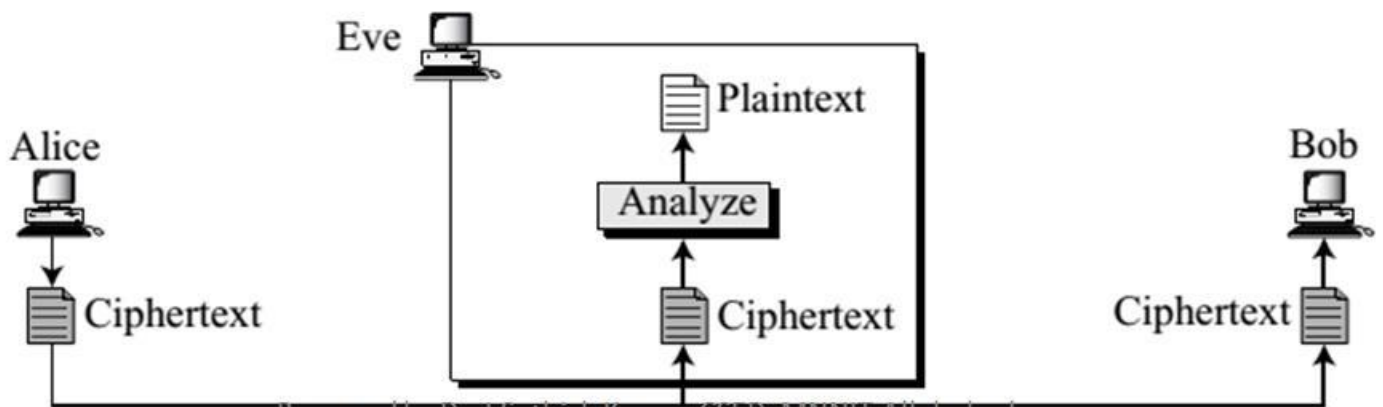
- Although it may appear that a cipher would be more secure if we hide both the encryption/decryption algorithm and the secret key, this is not recommended.
- Based on Kerckhoff's principle, one should always assume that the adversary, Eve, knows the encryption/decryption algorithm.
- The resistance of the cipher to attack must be based only on the secrecy of the key.
- In other words, guessing the key should be so difficult that there is no need to hide the encryption/decryption algorithm.
- This principle manifests itself more clearly when we study modern ciphers.
- There are only a few algorithms for modern ciphers today.
- The key domain for each algorithm, however, is so large that it makes it difficult for the adversary to find the key.

# Cryptanalysis



# Ciphertext-Only Attack

- In a ciphertext-only attack, Eve has access to only some ciphertext.
- She tries to find the corresponding key and the plaintext.
- The assumption is that Eve knows the algorithm and can intercept the ciphertext.
- The ciphertext-only attack is the most probable one because Eve needs only the ciphertext for this attack.
- To thwart the decryption of a message by an adversary, a cipher must be very resisting to this type of attack.

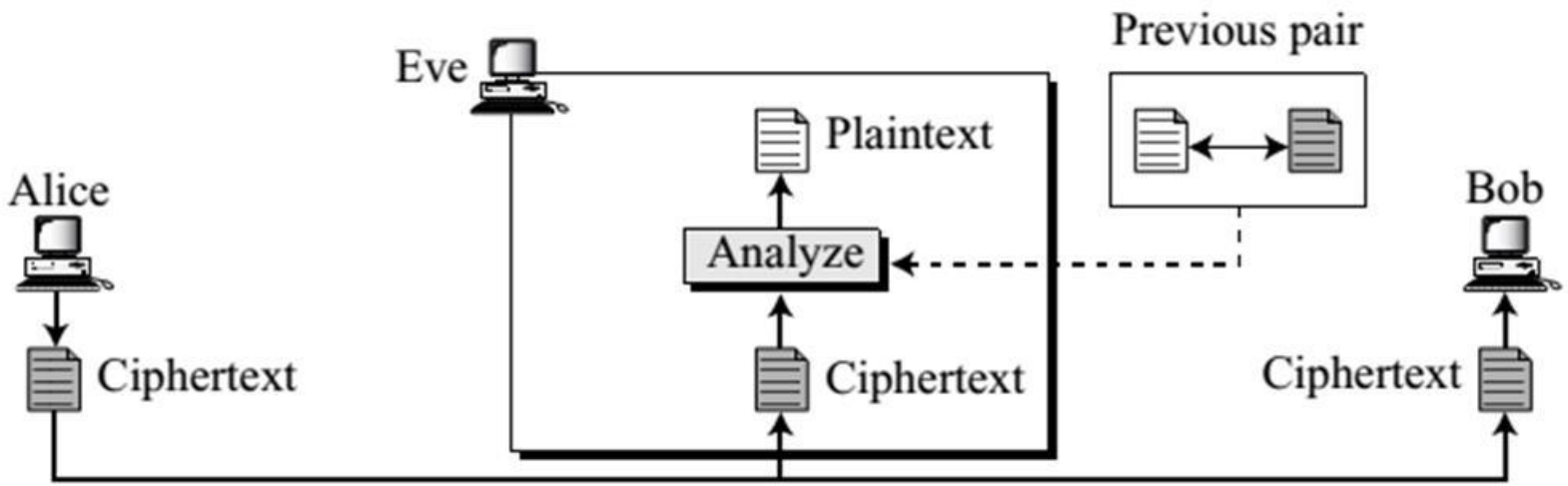


# Ciphertext-Only Attack

- Brute-Force Attack
  - In the brute-force method or exhaustive-key-search method, Eve tries to use all possible keys.
  - We assume that Eve knows the algorithm and knows the key domain (the list of all possible keys).
  - Using the intercepted cipher, Eve decrypts the ciphertext with every possible key until the plaintext makes sense.
  - Using brute-force attack was a difficult task in the past; it is easier today using a computer.
  - To prevent this type of attack, the number of possible keys must be very large.
- Statistical Attack
  - The cryptanalyst can benefit from some inherent characteristics of the plaintext language to launch a statistical attack.
  - For example, we know that the letter E is the most frequently used letter in English text.
  - The cryptanalyst finds the mostly-used character in the ciphertext and assumes that the corresponding plaintext character is E.
  - After finding a few pairs, the analyst can find the key and use it to decrypt the message.
  - To prevent this type of attack, the cipher should hide the characteristics of the language.
- Pattern Attack
  - Some ciphers may hide the characteristics of the language, but may create some patterns in the ciphertext.
  - A cryptanalyst may use a pattern attack to break the cipher.
  - Therefore, it is important to use ciphers that make the ciphertext look as random as possible.

# Known-Plaintext Attack

- In a known-plaintext attack, Eve has access to some plaintext/ciphertext pairs in addition to the intercepted ciphertext that she wants to break.

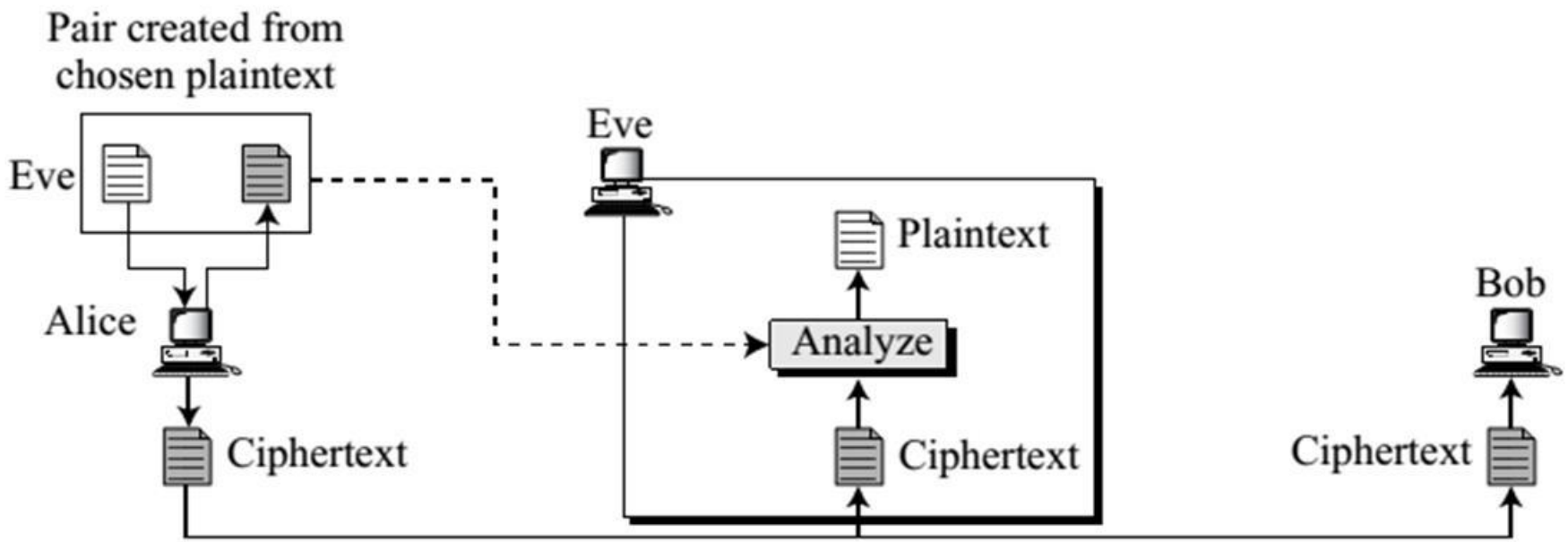


# Known-Plaintext Attack

- The plaintext/ciphertext pairs have been collected earlier.
- For example, Alice has sent a secret message to Bob, but she has later made the contents of the message public.
- Eve has kept both the ciphertext and the plaintext to use them to break the next secret message from Alice to Bob, assuming that Alice has not changed her key.
- Eve uses the relationship between the previous pair to analyze the current ciphertext.
- The same methods used in a ciphertext-only attack can be applied here.
- This attack is easier to implement because Eve has more information to use for analysis.
- However, it is less likely to happen because Alice may have changed her key or may have not disclosed the contents of any previous messages.

# Chosen-Plaintext Attack

- The chosen-plaintext attack is similar to the known-plaintext attack, but the plaintext/ciphertext pairs have been chosen by the attacker herself.

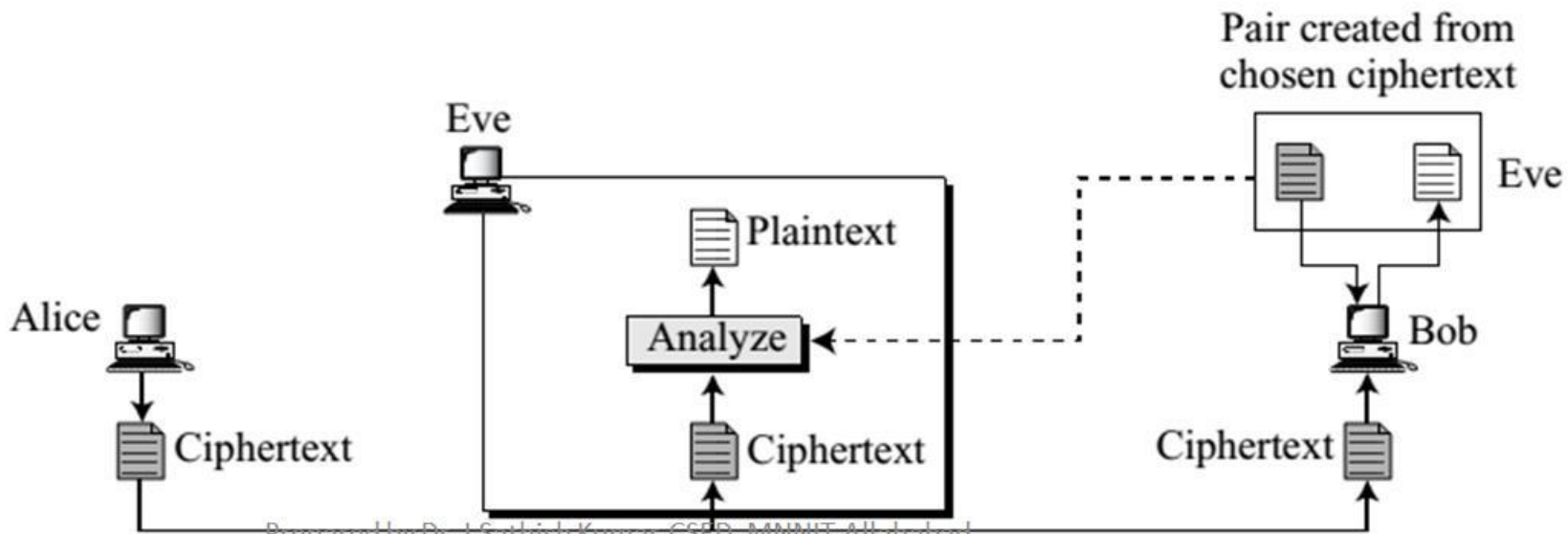


# Chosen-Plaintext Attack

- This can happen, for example, if Eve has access to Alice's computer.
- She can choose some plaintext and intercept the created ciphertext.
- Of course, she does not have the key because the key is normally embedded in the software used by the sender.
- This type of attack is much easier to implement, but it is much less likely to happen.

# Chosen-Ciphertext Attack

- The chosen-ciphertext attack is similar to the chosen-plaintext attack, except that Eve chooses some ciphertext and decrypts it to form a ciphertext/plaintext pair.
- This can happen if Eve has access to Bob's computer.



# Categories of Traditional Ciphers

- Substitution ciphers and Transposition ciphers.
- In a substitution cipher, we replace one symbol in the ciphertext with another symbol.
- In a transposition cipher, we reorder the position of symbols in the plaintext.

# SUBSTITUTION CIPHERS

- A substitution cipher replaces one symbol with another.
- Monoalphabetic Ciphers
  - In monoalphabetic substitution, the relationship between a symbol in the plaintext to a symbol in the ciphertext is always one-to-one.
  -

**Plaintext:** hello **Ciphertext:** KHOOR

**Plaintext:** hello **Ciphertext:** ABNZF

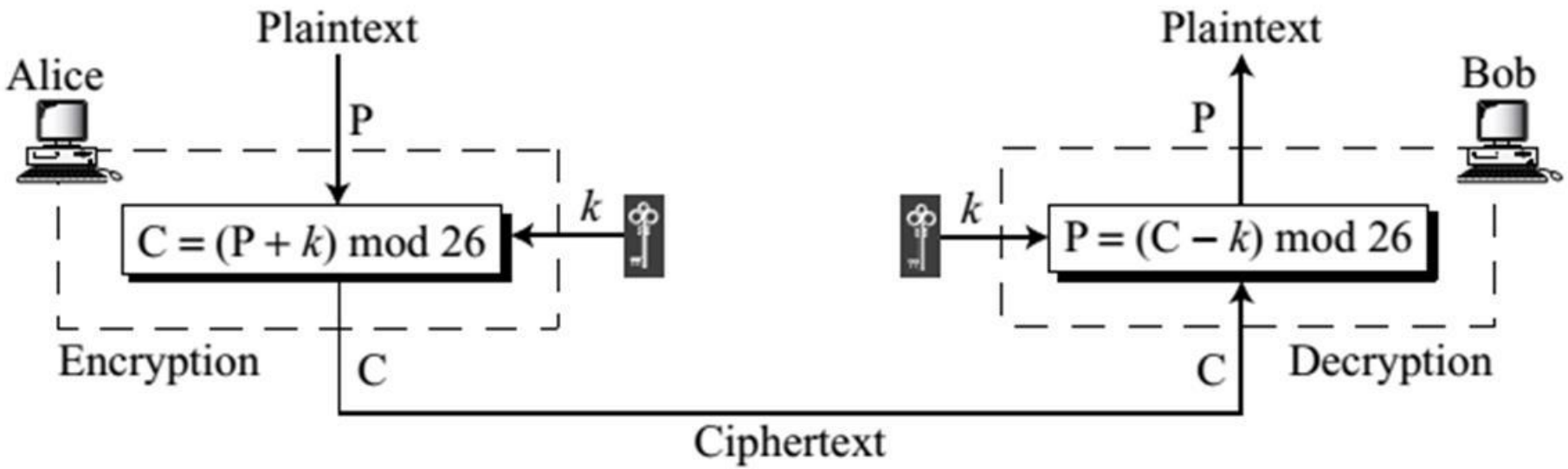
# Additive Cipher

- This cipher is sometimes called a shift cipher and sometimes a Caesar cipher, but the term additive cipher better reveals its mathematical nature.
- When the cipher is additive, the plaintext, ciphertext, and key are integers in $Z_{26}$.

| | | | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plaintext → | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
| Ciphertext → | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| Value → | 00 | 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 | 21 | 22 | 23 | 24 | 25 |

***Representation of plaintext and ciphertext characters in $Z_{26}$***

# *Additive Cipher*




$$P_1 = (C - k) \bmod 26 = (P + k - k) \text{ mode } 26 = P$$

# Example #1:
# Use the additive cipher with key = 15 to encrypt the message "hello". And decrypt to verify using the same key

| | | |
|---|---|---|
| Plaintext: h → 07 | Encryption: (07 + 15) mod 26 | Ciphertext: 22 → W |
| Plaintext: e → 04 | Encryption: (04 + 15) mod 26 | Ciphertext: 19 → T |
| Plaintext: l → 11 | Encryption: (11 + 15) mod 26 | Ciphertext: 00 → A |
| Plaintext: l → 11 | Encryption: (11 + 15) mod 26 | Ciphertext: 00 → A |
| Plaintext: o → 14 | Encryption: (14 + 15) mod 26 | Ciphertext: 03 → D |

| | | |
|---|---|---|
| Ciphertext: W → 22 | Decryption: (22 – 15) mod 26 | Plaintext: 07 → h |
| Ciphertext: T → 19 | Decryption: (19 – 15) mod 26 | Plaintext: 04 → e |
| Ciphertext: A → 00 | Decryption: (00 – 15) mod 26 | Plaintext: 11 → l |
| Ciphertext: A → 00 | Decryption: (00 – 15) mod 26 | Plaintext: 11 → l |
| Ciphertext: D → 03 | Decryption: (03 – 15) mod 26 | Plaintext: 14 → o |

- Additive ciphers are vulnerable to ciphertext-only attacks using exhaustive key searches (brute-force attacks).
- The key domain of the additive cipher is very small;there are only 26 keys.
- However, one of the keys, zero, is useless (the ciphertext is the same as the plaintext).
- This leaves only 25 possible keys.
- Eve can easily launch a bruteforce attack on the ciphertext.

s

# Example #2

Eve has intercepted the ciphertext "UVACLYFZLJBYL". Show how she can use a brute-force attack to break the cipher.

**Ciphertext:** UVACLYFZLJBYL

**K = 1** → **Plaintext:** tuzbkxeykiaxk
**K = 2** → **Plaintext:** styajwdxjhzwj
**K = 3** → **Plaintext:** rsxzivcwigyvi
**K = 4** → **Plaintext:** qrwyhubvhfxuh
**K = 5** → **Plaintext:** pqvxgtaugewtg
**K = 6** → **Plaintext:** opuwfsztfdvsf
**K = 7** → **Plaintext:** notverysecure

# Cryptanalysis

- Additive ciphers are also subject to statistical attacks. This is especially true if the adversary has a long ciphertext.
- The adversary can use the frequency of occurrence of characters for a particular language.

| *Letter* | *Frequency* | *Letter* | *Frequency* | *Letter* | *Frequency* | *Letter* | *Frequency* |
|---|---|---|---|---|---|---|---|
| E | 12.7 | H | 6.1 | W | 2.3 | K | 0.08 |
| T | 9.1 | R | 6.0 | F | 2.2 | J | 0.02 |
| A | 8.2 | D | 4.3 | G | 2.0 | Q | 0.01 |
| O | 7.5 | L | 4.0 | Y | 2.0 | X | 0.01 |
| I | 7.0 | C | 2.8 | P | 1.9 | Z | 0.01 |
| N | 6.7 | U | 2.8 | B | 1.5 | | |
| S | 6.3 | M | 2.4 | V | 1.0 | | |

# Cryptanalysis

| | |
|---|---|
| Digram | TH, HE, IN, ER, AN, RE, ED, ON, ES, ST, EN, AT, TO, NT, HA, ND, OU, EA, NG, AS, OR, TI, IS, ET, IT, AR, TE, SE, HI, OF |
| Trigram | THE, ING, AND, HER, ERE, ENT, THA, NTH, WAS, ETH, FOR, DTH |

The most common two-letter groups (digrams) and three-letter groups (trigrams) for the English text are shown in Table

# Example #3

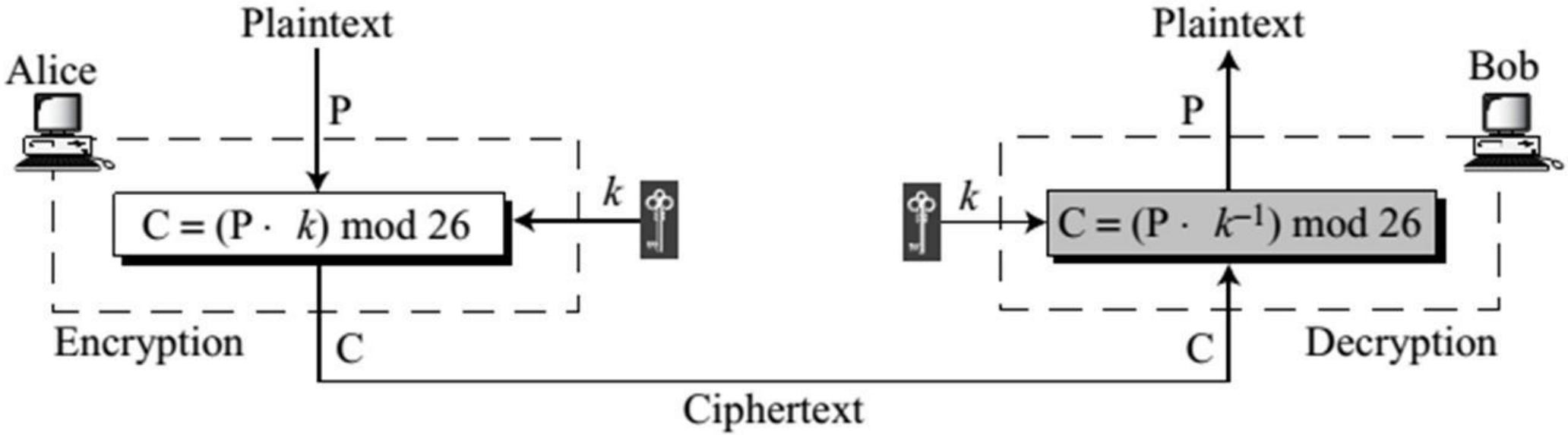
- Eve has intercepted the following ciphertext. Using a statistical attack, find the plaintext.

XLILSYWIMWRSAJSVWEPIJSVJSYVQMPPMSRHSPPEVWMXMWASVX-LQSVILY-VVCFIJSVIXLIWIPPIVVIGIMZIWQSVISJJIVW

- When Eve tabulates the frequency of letters in this ciphertext, she gets: I =14, V =13, S =12, and so on.
- The most common character is I with 14 occurrences.
- This shows that character I in the ciphertext probably corresponds to the character e in plaintext. This means key = 4. Eve deciphers the text to get

the house is now for sale for four million dollars it is worth more hurry before the seller receives more offers

# Multiplicative Ciphers




**In a multiplicative cipher, the plaintext and ciphertext are integers in Z26; The key is an integer in Z26*.**

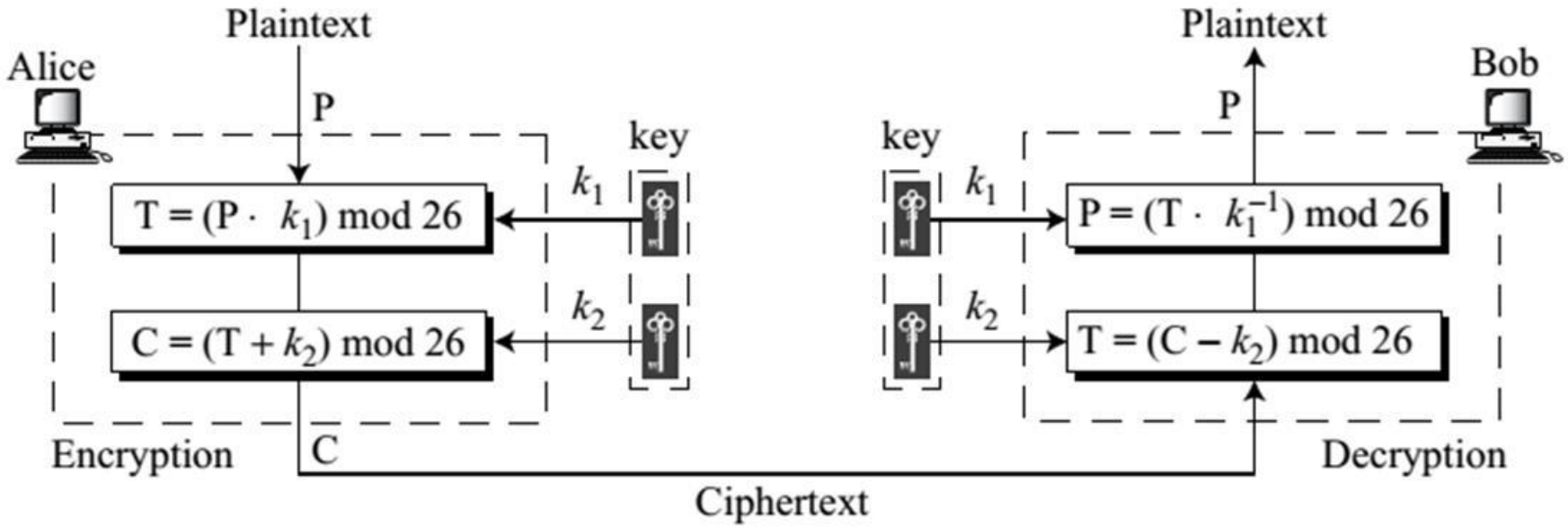
# Multiplicative Ciphers

- What is the key domain for any multiplicative cipher?
  - The key needs to be in $Z_{26}^*$. This set has only 12 members: 1, 3, 5, 7, 9, 11, 15, 17, 19, 21,23, 25.
- We use a multiplicative cipher to encrypt the message “hello” with a key of 7.

| | | |
|---|---|---|
| Plaintext: h $\rightarrow$ 07 | Encryption: $(07 \times 07) \bmod 26$ | ciphertext: 23 $\rightarrow$ X |
| Plaintext: e $\rightarrow$ 04 | Encryption: $(04 \times 07) \bmod 26$ | ciphertext: 02 $\rightarrow$ C |
| Plaintext: l $\rightarrow$ 11 | Encryption: $(11 \times 07) \bmod 26$ | ciphertext: 25 $\rightarrow$ Z |
| Plaintext: l $\rightarrow$ 11 | Encryption: $(11 \times 07) \bmod 26$ | ciphertext: 25 $\rightarrow$ Z |
| Plaintext: o $\rightarrow$ 14 | Encryption: $(14 \times 07) \bmod 26$ | ciphertext: 20 $\rightarrow$ U |

# Affine Cipher

- Combine the additive and multiplicative ciphers to get what is called the affine cipher-a combination of both ciphers with a pair of keys.
- The first key is used with the multiplicative cipher; the second key is used with the additive cipher.



# Affine Cipher

$$C = (P \times k_1 + k_2) \bmod 26 \qquad P = ((C - k_2) \times k_1^{-1}) \bmod 26$$

where $k_1^{-1}$ is the multiplicative inverse of $k_1$ and $-k_2$ is the additive inverse of $k_2$

Use an affine cipher to encrypt the message "hello" with the key pair (7, 2).

| | | |
|---|---|---|
| P: h → 07 | Encryption: $(07 \times 7 + 2) \bmod 26$ | C: 25 → Z |
| P: e → 04 | Encryption: $(04 \times 7 + 2) \bmod 26$ | C: 04 → E |
| P: l → 11 | Encryption: $(11 \times 7 + 2) \bmod 26$ | C: 01 → B |
| P: l → 11 | Encryption: $(11 \times 7 + 2) \bmod 26$ | C: 01 → B |
| P: o → 14 | Encryption: $(14 \times 7 + 2) \bmod 26$ | C: 22 → W |

# Affine Cipher

- Use the affine cipher to decrypt the message “ZEBBW” with the key pair (7, 2) in modulus 26.
- Add the additive inverse of $-2 \equiv 24 \pmod{26}$ to the received ciphertext.
- Then multiply the result by the multiplicative inverse of $7^{-1} \equiv 15 \pmod{26}$ to find the plaintext characters.
- Because 2 has an additive inverse in $Z_{26}$ and 7 has a multiplicative inverse in $Z_{26}^*$

| | | |
|---|---|---|
| C: Z $\rightarrow$ 25 | Decryption: $((25-2)\times 7^{-1}) \bmod 26$ | P:07 $\rightarrow$ h |
| C: E $\rightarrow$ 04 | Decryption: $((04-2)\times 7^{-1}) \bmod 26$ | P:04 $\rightarrow$ e |
| C: B $\rightarrow$ 01 | Decryption: $((01-2)\times 7^{-1}) \bmod 26$ | P:11 $\rightarrow$ l |
| C: B $\rightarrow$ 01 | Decryption: $((01-2)\times 7^{-1}) \bmod 26$ | P:11 $\rightarrow$ l |
| C: W $\rightarrow$ 22 | Decryption: $((22-2)\times 7^{-1}) \bmod 26$ | P:14 $\rightarrow$ o |

# Cryptanalysis of Affine Cipher

PWUFFOGWCHFDWIWEJOUUNJORSMDWRHVCMWJUPVCCG

Although the brute-force and statistical method of ciphertext-only attack can be used, let us try a chosen-plaintext attack.

Eve also very briefly obtains access to Alice's computer and has only enough time to type a two-letter plaintext: "et".

She then tries to encrypt the short plaintext using two different algorithms, because she is not sure which one is the affine cipher.

| | | |
|---|---|---|
| Algorithm 1: | Plaintext: et | ciphertext: → WC |
| Algorithm 2: | Plaintext: et | ciphertext: → WF |

# Cryptanalysis of Affine Cipher

Eve knows that if the first algorithm is affine, she can construct the following two equations based on the first data set.

| | | |
|---|---|---|
| e → W | 04 → 22 | $(04 \times k_1 + k_2) \equiv 22 \pmod{26}$ |
| t → C | 19 → 02 | $(19 \times k_1 + k_2) \equiv 02 \pmod{26}$ |

These two congruence equations can be solved and the values of $k_1$ and $k_2$ can be found.

$$\begin{bmatrix} k_1 \\ k_2 \end{bmatrix} = \begin{bmatrix} 4 & 1 \\ 19 & 1 \end{bmatrix}^{-1} \begin{bmatrix} 22 \\ 2 \end{bmatrix} = \begin{bmatrix} 19 & 7 \\ 3 & 24 \end{bmatrix} \begin{bmatrix} 22 \\ 2 \end{bmatrix} = \begin{bmatrix} 16 \\ 10 \end{bmatrix} \longrightarrow k_1 = 16 \quad k_2 = 10$$

However, this answer is not acceptable because $k_1 = 16$ cannot be the first part of the key. Its value, 16, does not have a multiplicative inverse in $Z_{26}^*$.

# Cryptanalysis of Affine Cipher

Eve now tries the result of the second set of data

| | | |
|---|---|---|
| e → W | 04 → 22 | $(04 \times k_1 + k_2) \equiv 22 \pmod{26}$ |
| t → F | 19 → 05 | $(19 \times k_1 + k_2) \equiv 05 \pmod{26}$ |

The square matrix and its inverse are the same. Now she has $k_1 = 11$ and $k_2 = 4$.
This pair is acceptable because $k_1$ has a multiplicative inverse in $Z_{26}^*$.
She tries the pair of keys (19, 22), which are the inverse of the pair (11, 4), to decipher the message. The plaintext is

best time of the year is spring when flowers bloom

# Monoalphabetic Substitution Cipher

- In this method, it is to create a mapping between each plaintext character and the corresponding ciphertext character.
- Alice and Bob can agree on a table showing the mapping for each character.

| Plaintext → | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Ciphertext → | N | O | A | T | R | B | E | C | F | U | X | D | Q | G | Y | L | K | H | V | I | J | M | P | Z | S | W |

An example key for monoalphabetic substitution cipher

this message is easy to encrypt but hard to find the key

ICFVQRVVNEFVRNVSIYRGAHSLIOJICNHTIYBFGTICRXRS

# Cryptanalysis

- The size of the key space for the monoalphabetic substitution cipher is 26! .
- This makes a brute-force attack extremely difficult for Eve even if she is using a powerful computer.
- However, she can use statistical attack based on the frequency of characters.
- The cipher does not change the frequency of characters.
- The monoalphabetic ciphers do not change the frequency of characters in the ciphertext, which makes the ciphers vulnerable to statistical attack.

# Polyalphabetic Ciphers

- In polyalphabetic substitution, each occurrence of a character may have a different substitute.
- The relationship between a character in the plaintext to a character in the ciphertext is one-to-many.
- Polyalphabetic ciphers have the advantage of hiding the letter frequency of the underlying language.
- Eve cannot use single-letter frequency statistic to break the ciphertext.

# Autokey Cipher

- In this cipher, the key is a stream of subkeys, in which each subkey is used to encrypt the corresponding character in the plaintext.
- The first subkey is a predetermined value secretly agreed upon by Alice and Bob.
- The second subkey is the value of the first plaintext character (between 0 and 25).
- The third subkey is the value of the second plaintext. And so on.

$P = P_1P_2P_3 \ldots$ $C = C_1C_2C_3\ldots$ $k = (k_1, P_1, P_2, \ldots)$

Encryption: $C_i = (P_i + k_i) \bmod 26$ Decryption: $P_i = (C_i - k_i) \bmod 26$

# Autokey Cipher

- However, it is still as vulnerable to the brute-force attack as the additive cipher.

| | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| Plaintext: | a | t | t | a | c | k | i | s | t | o | d | a | y |
| P's Values: | 00 | 19 | 19 | 00 | 02 | 10 | 08 | 18 | 19 | 14 | 03 | 00 | 24 |
| Key stream: | *12* | *00* | *19* | *19* | *00* | *02* | *10* | *08* | *18* | *19* | *14* | *03* | *00* |
| C's Values: | 12 | 19 | 12 | 19 | 02 | 12 | 18 | 00 | 11 | 7 | 17 | 03 | 24 |
| Ciphertext: | **M** | **T** | **M** | **T** | **C** | **M** | **S** | **A** | **L** | **H** | **R** | **D** | **Y** |

However, it is still as vulnerable to the brute-force attack as the additive cipher.

# Playfair Cipher

- Playfair cipher used by the British army during World War I.
- The secret key in this cipher is made of 25 alphabet letters arranged in a 5 × 5 matrix (letters I and J are considered the same when encrypting).
- Different arrangements of the letters in the matrix can create many different secret keys.
- The letters dropped in the matrix diagonally starting from the top right-hand corner.
- Before encryption,
  - if the two letters in a pair are the same, a bogus letter is inserted to separate them.
  - After inserting bogus letters, if the number of characters in the plaintext is odd, one extra bogus character is added at the end to make the number of characters even.

**Secret Key =**

| | | | | |
|---|---|---|---|---|
| L | G | D | B | A |
| Q | M | H | E | C |
| U | R | N | I/J | F |
| X | V | S | O | K |
| Z | Y | W | T | P |

# Playfair Cipher

- The cipher uses three rules for encryption:

  a. If the two letters in a pair are located in the same row of the secret key, the corresponding encrypted character for each letter is the next letter to the right in the same row (with wrapping to the beginning of the row if the plaintext letter is the last character in the row).

  b. If the two letters in a pair are located in the same column of the secret key, the corresponding encrypted character for each letter is the letter beneath it in the same column (with wrapping to the beginning of the column if the plaintext letter is the last character in the column).

  c. If the two letters in a pair are not in the same row or column of the secret, the corresponding encrypted character for each letter is a letter that is in its own row but in the same column as the other letter.

# Playfair Cipher

$P = P_1P_2P_3 \ldots$ $C = C_1C_2C_3 \ldots$ $k = [(k_1, k_2), (k_3, k_4), \ldots]$

Encryption: $C_i = k_i$ Decryption: $P_i = k_i$

Let us encrypt the plaintext "hello" using the key in Figure

**Secret Key =**

| | | | | |
|---|---|---|---|---|
| L | G | D | B | A |
| Q | M | H | E | C |
| U | R | N | I/J | F |
| X | V | S | O | K |
| Z | Y | W | T | P |

When we group the letters in two-character pairs, we get "he, ll, o".
We need to insert an x between the two l's (els), giving "he, lx, lo".

he → EC lx → QZ lo → BX

Plaintext: hello Ciphertext: ECQZBX

# Cryptanalysis of a Playfair Cipher

- Obviously a brute-force attack on a Playfair cipher is very difficult.
  - The size of the key domain is 25!
- However, the frequencies of diagrams are preserved (to some extent because of filler insertion), so a cryptanalyst can use a ciphertext-only attack based on the digram frequency test to find the key.

# Vigenere Cipher

- A Vigenere cipher uses a different strategy to create the key stream.
- The key stream is a repetition of an initial secret key stream of length m, where we have $1 \leq m \leq 26$.
- The cipher can be described as follows where (k1,k2, …, km) is the initial secret key agreed to by Alice and Bob.

$$P = P_1P_2P_3 \ldots \qquad C = C_1C_2C_3 \ldots \qquad K = [(k_1, k_2, \ldots, k_m), (k_1, k_2, \ldots, k_m), \ldots]$$

$$\text{Encryption: } C_i = P_i + k_i \qquad \text{Decryption: } P_i = C_i - k_i$$

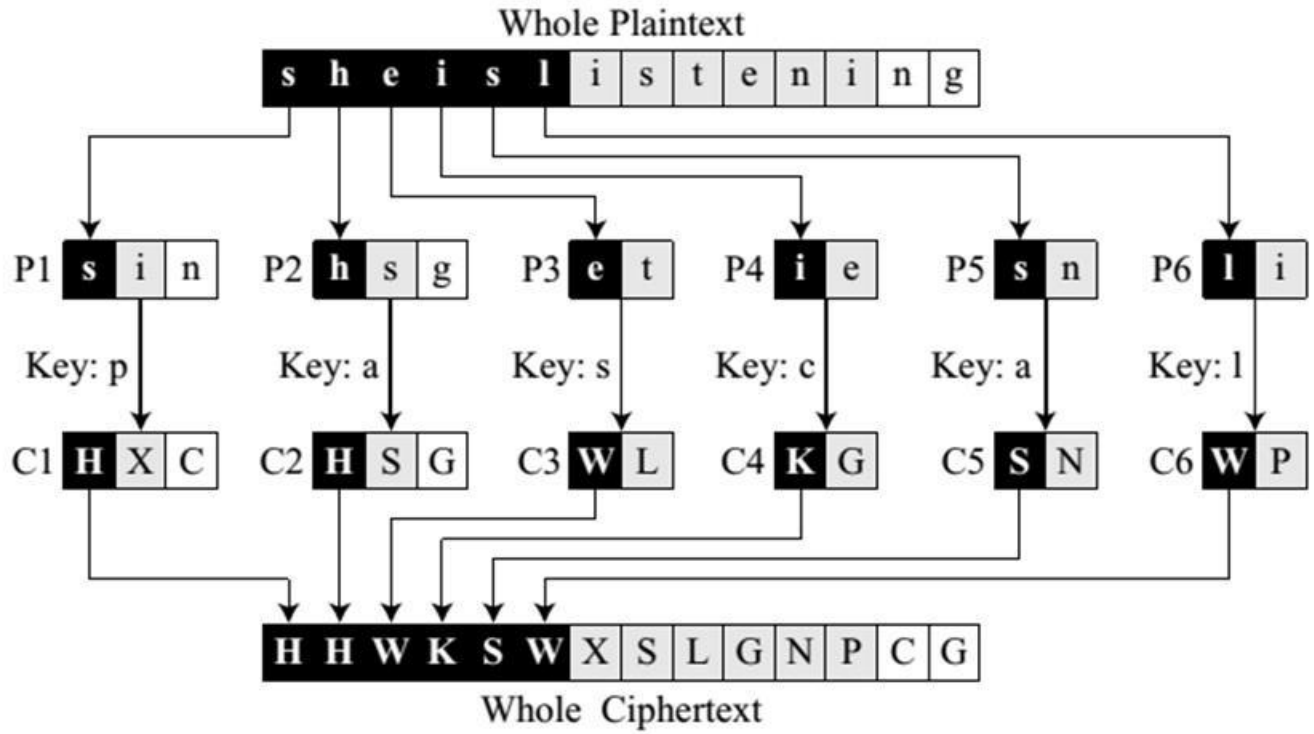
# Vigenere Cipher

- Let us see how we can encrypt the message “She is listening” using the 6-character keyword “PASCAL”.
- The initial key stream is (15, 0, 18, 2, 0, 11).
- The key stream is the repetition of this initial key stream (as many times as needed).

| **Plaintext:** | s | h | e | i | s | l | i | s | t | e | n | i | n | g |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **P’s values:** | 18 | 07 | 04 | 08 | 18 | 11 | 08 | 18 | 19 | 04 | 13 | 08 | 13 | 06 |
| **Key stream:** | ***15*** | ***00*** | ***18*** | ***02*** | ***00*** | ***11*** | ***15*** | ***00*** | ***18*** | ***02*** | ***00*** | ***11*** | ***15*** | ***00*** |
| **C’s values:** | **07** | **07** | **22** | **10** | **18** | **22** | **23** | **18** | **11** | **6** | **13** | **19** | **02** | **06** |
| **Ciphertext:** | **H** | **H** | **W** | **K** | **S** | **W** | **X** | **S** | **L** | **G** | **N** | **T** | **C** | **G** |

# Vigenere Cipher

- Vigenere cipher can be seen as combinations of m additive ciphers.
- The figure shows how the plaintext of the previous example can be thought of as six different pieces, each encrypted separately.
- The figure helps us later understand the cryptanalysis of Vigenere ciphers.
- There are m pieces of the plaintext, each encrypted with a different key, to make m pieces of ciphertext.



# Vigenere Cipher

Another way to look at Vigenere Ciphers is through what is called a Vigenere tableau

The additive cipher is a special case of Vigenere cipher in which m = 1.

| | a | b | c | d | e | f | g | h | i | j | k | l | m | n | o | p | q | r | s | t | u | v | w | x | y | z |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| *A* | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z |
| *B* | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A |
| *C* | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B |
| *D* | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C |
| *E* | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D |
| *F* | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E |
| *G* | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F |
| *H* | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G |
| *I* | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H |
| *J* | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I |
| *K* | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J |
| *L* | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K |
| *M* | M | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L |
| *N* | N | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M |
| *O* | O | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N |
| *P* | P | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O |
| *Q* | Q | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P |
| *R* | R | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q |
| *S* | S | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R |
| *T* | T | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S |
| *U* | U | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T |
| *V* | V | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U |
| *W* | W | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V |
| *X* | X | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W |
| *Y* | Y | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X |
| *Z* | Z | A | B | C | D | E | F | G | H | I | J | K | L | M | N | O | P | Q | R | S | T | U | V | W | X | Y |

# Cryptanalysis of Vigenere Ciphers

- Vigenere ciphers, like all polyalphabetic ciphers, do not preserve the frequency of characters.
- However, cryptanalysis here consists of two parts:
  - finding the length of the key and finding the key itself.
- Kasiski test
  - Searches for repeated text segments, of at least three characters, in the ciphertext.
  - Suppose that two of these segments are found and the distance between them is d.
  - The cryptanalyst assumes that d | m where m is the key length.
  - If more repeated segments can be found with distances $d_1$, $d_2$, …, $d_n$, then gcd ($d_1$, $d_2$, …, $d_n$)/m.
- After the length of the key has been found, the cryptanalyst uses the idea m=1.
- She divides the ciphertext into m different pieces and applies the method used to cryptanalyze the additive cipher, including frequency attack.
- Each ciphertext piece can be decrypted and put together to create the whole plaintext.
- In other words, the whole ciphertext does not preserve the single-letter frequency of the plaintext, but each piece does.

# Cryptanalysis of Vigenere Ciphers

LIOMWGFEGGDVWGHHCQUCRHRWAGWIOWQLKGZETKKMEVLWPCZVGTH-
VTSGXQOVGCSVETQLTJSUMVWVEUVLXEWSLGFZMVVWLGYHCUSWXQH-
KVGSHEEVFLCFDGVSUMPHKIRZDMPHHBVWVWJWIXGFWLTSHGJOUEEHH-
VUCFVGOWICQLTJSUXGLW

The Kasiski test for repetition of three-character segments yields the results

| *String* | *First Index* | *Second Index* | *Difference* |
|---|---|---|---|
| JSU | 68 | 168 | 100 |
| SUM | 69 | 117 | 48 |
| VWV | 72 | 132 | 60 |
| MPH | 119 | 127 | 8 |

# Cryptanalysis of Vigenere Ciphers

- The greatest common divisor of differences is 4, which means that the key length is multiple of 4.
- First try m = 4.
- Divide the ciphertext into four pieces.
- Piece C1 is made of characters 1, 5, 9,…;
- piece C2 is made of characters 2, 6, 10, …; and so on.
- Use the statistical attack on each piece separately.
- Interleave the decipher pieces one character at a time to get the whole plaintext.
- If the plaintext does not make sense, try with another m.

# Cryptanalysis of Vigenere Ciphers

```
C1: LWGWCRAOKTEPGTQCTJVUEGVGUQGECVPRPVJGTJEUGCJG
P1: jueuapymircneroarhtsthihytrahcieixsthcarrehe
C2: IGGGQHGWGKVCTSOSQSWVWFVYSHSVFSHZHWWFSOHCOQSL
P2: usssctsiswhofeaeceihcetesoecatnpntherhctecex
C3: OFDHURWQZKLZHGVVLUVLSZWHWKHFDUKDHVIWHUHFWLUW
P3: lcaerotnwhiwedssirsiirhketehretltiideatrairt
C4: MEVHCWILEMWVVXGETMEXLMLCXVELGMIMBWXLGEVVITX
P4: iardysehaisrrtcapiafpwtethecarhaesfterectpt
```

**Julius Caesar used a cryptosystem in his wars, which is now referred to as Caesar cipher. It is an additive cipher with the key set to three. Each character in the plaintext is shifted three characters to create ciphertext.**

# Hill Cipher

- The plaintext is divided into equal-size blocks.
- The blocks are encrypted one at a time in such a way that each character in the block contributes to the encryption of other characters in the block.
- For this reason, the Hill cipher belongs to a category of ciphers called block ciphers.
- The other ciphers we studied so far belong to the category called stream ciphers.
- In a Hill cipher, the key is a square matrix of size m × m in which m is the size of the block.
- If we call the key matrix K, each element of the matrix is $k_{i,j}$ as shown in Figure

$$\mathbf{K} = \begin{bmatrix} k_{11} & k_{12} & \cdots & k_{1m} \\ k_{21} & k_{22} & \cdots & k_{2m} \\ \vdots & \vdots & & \vdots \\ k_{m1} & k_{m2} & \cdots & k_{mm} \end{bmatrix}$$

# Hill Cipher

Let us show how one block of the ciphertext is encrypted.
If we call the m characters in the plaintext block P1, P2, …, Pm, the corresponding characters in the ciphertext block are C1, C2, …, Cm.

$$
\begin{aligned}
C_1 &= P_1 k_{11} + P_2 k_{21} + \cdots + P_m k_{m1} \\
C_2 &= P_1 k_{12} + P_2 k_{22} + \cdots + P_m k_{m2} \\
&\cdots \\
C_m &= P_1 k_{1m} + P_2 k_{2m} + \cdots + P_m k_{mm}
\end{aligned}
$$

The key matrix in the Hill Cipher needs to have a multiplicative inverse.

The equations show that each ciphertext character such as C1 depends on all plaintext characters in the block (P1, P2, …, Pm).
However, we should be aware that not all square matrices have multiplicative inverses in Z26, so Alice and Bob should be careful in selecting the key.
Bob will not be able to decrypt the ciphertext sent by Alice if the matrix does not have a multiplicative inverse.

# Hill Cipher

- For example, the plaintext "code is ready" can make a 3 × 4 matrix when adding extra bogus character "z" to the last block and removing the spaces.
- The ciphertext is "OHKNIHGKLISS".
- Bob can decrypt the message using the inverse of the key matrix.

$$\underset{\mathbf{C}}{\begin{bmatrix} 14 & 07 & 10 & 13 \\ 08 & 07 & 06 & 11 \\ 11 & 08 & 18 & 18 \end{bmatrix}} = \underset{\mathbf{P}}{\begin{bmatrix} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{bmatrix}} \overset{\mathbf{K}}{\begin{bmatrix} 09 & 07 & 11 & 13 \\ 04 & 07 & 05 & 06 \\ 02 & 21 & 14 & 09 \\ 03 & 23 & 21 & 08 \end{bmatrix}}$$

a. Encryption

$$\underset{\mathbf{P}}{\begin{bmatrix} 02 & 14 & 03 & 04 \\ 08 & 18 & 17 & 04 \\ 00 & 03 & 24 & 25 \end{bmatrix}} = \underset{\mathbf{C}}{\begin{bmatrix} 14 & 07 & 10 & 13 \\ 08 & 07 & 06 & 11 \\ 11 & 08 & 18 & 18 \end{bmatrix}} \overset{\mathbf{K}^{-1}}{\begin{bmatrix} 02 & 15 & 22 & 03 \\ 15 & 00 & 19 & 03 \\ 09 & 09 & 03 & 11 \\ 17 & 00 & 04 & 07 \end{bmatrix}}$$

b. Decryption

# Cryptanalysis of Hill Ciphers

- A brute-force attack on a Hill cipher is extremely difficult because the key is an m × m matrix.
  - Each entry in the matrix can have one of the 26 values.
  - This means that the size of the key domain is $26^{m \times m}$
- Hill ciphers do not preserve the statistics of the plaintext.
  - Eve cannot run frequency analysis on single letters, digrams, or trigrams.
  - A frequency analysis of words of size m might work, but this is very rare that a plaintext has many strings of size m that are the same.

# Cryptanalysis of Hill Ciphers

- Eve, however, can do a known-plaintext attack on the cipher if she knows the value of m and knows the plaintext/ciphertext pairs for at least m blocks.
  - The blocks can belong to the same message or different messages but should be distinct.
  - Eve can create two m × m matrices, P (plaintext) and C (ciphertext) in which the corresponding rows represent the corresponding known plaintext/ciphertext pairs.
  - Because $C = PK$, Eve can use the relationship $K = CP^{-1}$ to find the key if P is invertible.
  - If P is not invertible, then Eve needs to use a different set of m plaintext/ciphertext pairs.
  - If Eve does not know the value of m, she can try different values provided that m is not very large.

# Cryptanalysis of Hill Ciphers

- Assume that Eve knows that m = 3. She has intercepted three plaintext/ciphertext pair blocks (not necessarily from the same message)
- She makes matrices P and C from these pairs. Because P is invertible, she inverts the P matrix and multiplies it by C to get the K matrix

$$\begin{bmatrix} 05 & 07 & 10 \end{bmatrix} \longleftrightarrow \begin{bmatrix} 03 & 06 & 00 \end{bmatrix}$$
$$\begin{bmatrix} 13 & 17 & 07 \end{bmatrix} \longleftrightarrow \begin{bmatrix} 14 & 16 & 09 \end{bmatrix}$$
$$\begin{bmatrix} 00 & 05 & 04 \end{bmatrix} \longleftrightarrow \begin{bmatrix} 03 & 17 & 11 \end{bmatrix}$$

$\mathbf{P}$ $\mathbf{C}$

$$\begin{bmatrix} 02 & 03 & 07 \\ 05 & 07 & 09 \\ 01 & 02 & 11 \end{bmatrix} = \begin{bmatrix} 21 & 14 & 01 \\ 00 & 08 & 25 \\ 13 & 03 & 08 \end{bmatrix} \begin{bmatrix} 03 & 06 & 00 \\ 14 & 16 & 09 \\ 03 & 17 & 11 \end{bmatrix}$$

$\mathbf{K}$ $\mathbf{P}^{-1}$ $\mathbf{C}$

*Now she has the key and can break any ciphertext encrypted with that key.*

# One-Time Pad

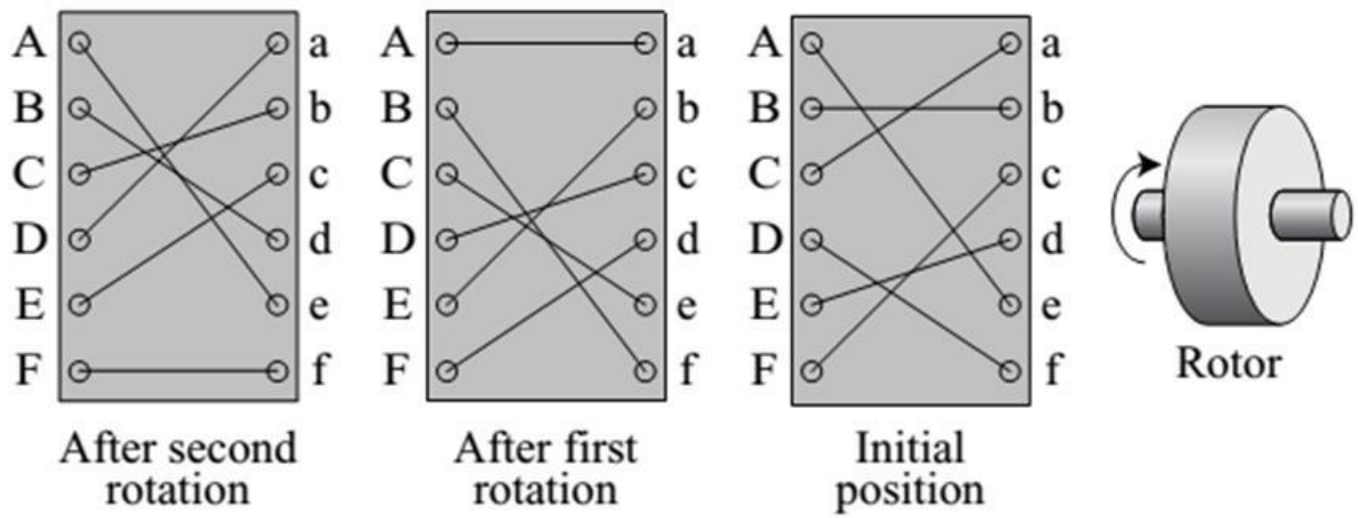
- One of the goals of cryptography is perfect secrecy.
- A study by Shannon has shown that perfect secrecy can be achieved if each plaintext symbol is encrypted with a key randomly chosen from a key domain.
- For example, an additive cipher can be easily broken because the same key is used to encrypt every character.
- However, even this simple cipher can become a perfect cipher if the key that is used to encrypt each character is chosen randomly from the key domain (00, 01, 02, …, 25) - that is, if the first character is encrypted using the key 04, the second character is encrypted using the key 02, the third character is encrypted using the key 21; and so on.
- This idea is used in a cipher called one-time pad, invented by Vernam.
- In this cipher, the key has the same length as the plaintext and is chosen completely in random.

# One-Time Pad

- A one-time pad is a perfect cipher, but it is almost impossible to implement commercially.
- If the key must be newly generated each time, how can Alice tell Bob the new key each time she has a message to send?
- However, there are some occasions when a one-time pad can be used.
- For example, if the president of a country needs to send a completely secret message to the president of another country, she can send a trusted envoy with the random key before sending the message.

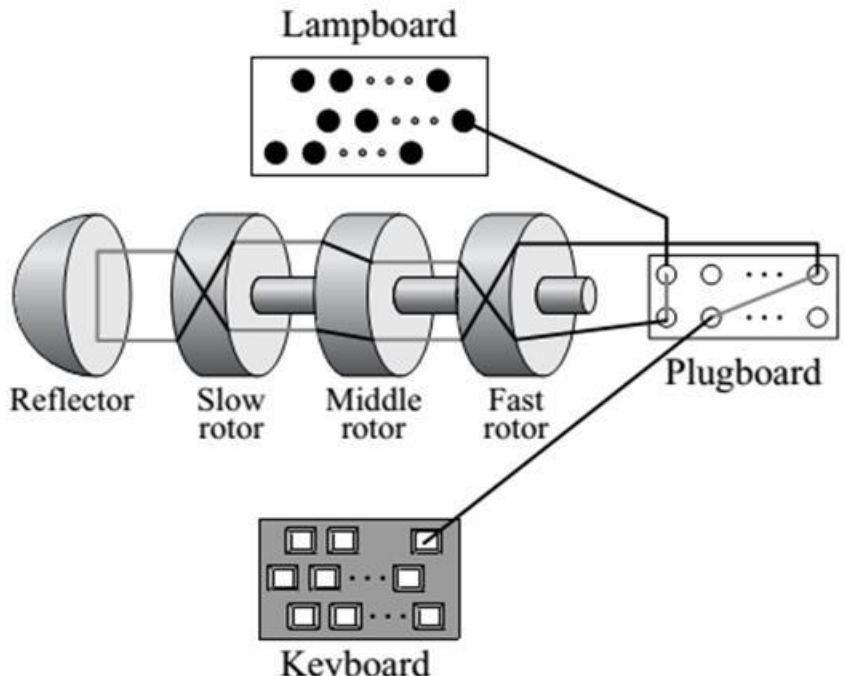
# Rotor Cipher

- Although one-time pad ciphers are not practical, one step toward more secured encipherment is the rotor cipher.
- It uses the idea behind monoalphabetic substitution but changes the mapping between the plaintext and the ciphertext characters for each plaintext character.



# Enigma Machine

- The Enigma machine was originally invented by Sherbius, but was modified by the German army and extensively used during World War II.
- The machine was based on the principle of rotor ciphers.

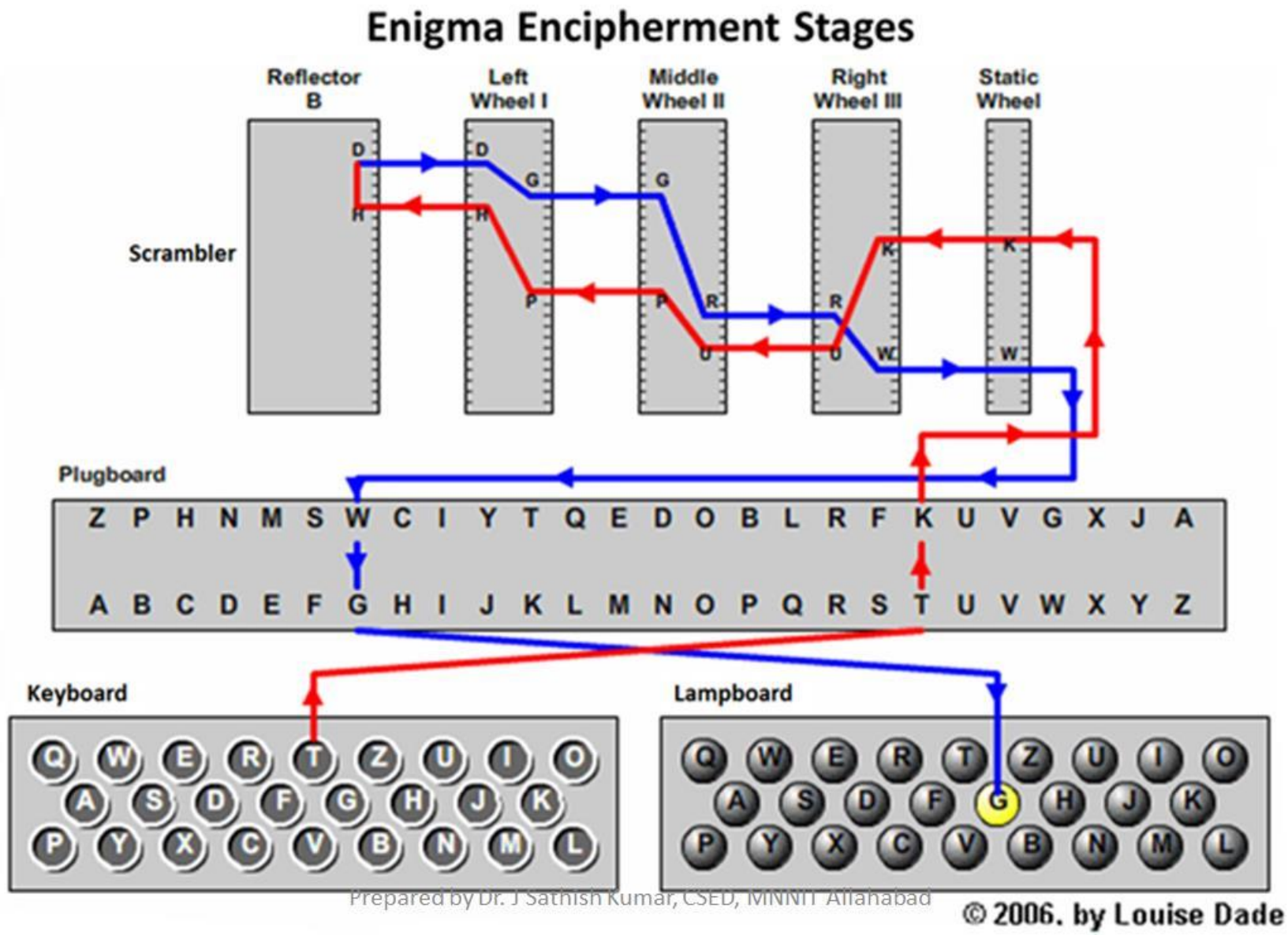


# Modern Cryptography

- 1918: The Enigma – Arthur Scherbius
- Business: confidential docs
- No codebooks
- Rotors -> multi substitution
- Wireing changes as-youtype
- German forces in WWII



# Modern Cryptography: The Enigma



# Categories of Traditional Ciphers

- Substitution ciphers and Transposition ciphers.
- In a substitution cipher, we replace one symbol in the ciphertext with another symbol.
- In a transposition cipher, we reorder the position of symbols in the plaintext.

# TRANSPOSITION CIPHERS

- A transposition cipher does not substitute one symbol for another, instead it changes the location of the symbols.
- A symbol in the first position of the plaintext may appear in the tenth position of the ciphertext.
- A symbol in the eighth position in the plaintext may appear in the first position of the ciphertext.
- In other words, a transposition cipher reorders (transposes) the symbols.

# Keyless Transposition Ciphers

- Simple transposition ciphers, which were used in the past, are keyless.
- There are two methods for permutation of characters.
- In the first method, the text is written into a table column by column and then transmitted row by row.
- In the second method, the text is written into the table row by row and then transmitted column by column.

# Keyless Transposition Ciphers
# Rail fence cipher

- In this cipher, the plaintext is arranged in two lines as a zigzag pattern (which means column by column);
- The ciphertext is created reading the pattern row by row.
- For example, to send the message “Meet me at the park” to Bob, Alice writes




- She then creates the ciphertext “MEMATEAKETETHPR” by sending the first row followed by the second row.
- Bob receives the ciphertext and divides it in half (in this case the second half has one less character).
- The first half forms the first row; the second half, the second row. Bob reads the result in zigzag.

# Keyless Transposition Ciphers
## Transposition cipher

- Alice and Bob can agree on the number of columns and use the second method.
- Alice writes the same plaintext, row by row, in a table of four columns.
- She then creates the ciphertext "MMTAEEHREAEKTTP" by transmitting the characters column by column.
- Bob receives the ciphertext and follows the reverse process.
- He writes the received message, column by column, and reads it row by row as the plaintext.

| | | | |
|---|---|---|---|
| m | e | e | t |
| m | e | a | t |
| t | h | e | p |
| a | r | k | |

# Keyless Transposition Ciphers
# Transposition cipher

- The following figure shows the permutation of each character in the plaintext into the ciphertext based on the positions.

| 01 | 02 | 03 | 04 | 05 | 06 | 07 | 08 | 09 | 10 | 11 | 12 | 13 | 14 | 15 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ | ↓ |
| 01 | 05 | 09 | 13 | 02 | 06 | 10 | 13 | 03 | 07 | 11 | 15 | 04 | 08 | 12 |

| | | | |
|---|---|---|---|
| **m** | **e** | **e** | **t** |
| **m** | **e** | **a** | **t** |
| **t** | **h** | **e** | **p** |
| **a** | **r** | **k** | |

- The second character in the plaintext has moved to the fifth position in the ciphertext;
- The third character has moved to the ninth position; and so on.
- Although the characters are permuted, there is a pattern in the permutation: (01, 05, 09, 13), (02, 06, 10, 13), (03, 07, 11, 15), and (08,12).
- In each section, the difference between the two adjacent numbers is 4.

# Keyed Transposition Ciphers

- The keyless ciphers permute the characters by using writing plaintext in one way (row by row, for example) and reading it in another way (column by column, for example).
- The permutation is done on the whole plaintext to create the whole ciphertext.
- Another method is to divide the plaintext into groups of predetermined size, called blocks, and then use a key to permute the characters in each block separately.

# Keyed Transposition Ciphers

- Alice needs to send the message "Enemy attacks tonight" to Bob.
- Alice and Bob have agreed to divide the text into groups of five characters and then permute the characters in each group.
- The following shows the grouping after adding a bogus character at the end to make the last group the same size as the others.

| | | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| e | n | e | m | y | | a | t | t | a | c | | k | s | t | o | n | | i | g | h | t | z |

- The key used for encryption and decryption is a permutation key, which shows how the character are permuted.
- For this message, assume that Alice and Bob used the following key

Encryption ↓

| | | | | |
|---|---|---|---|---|
| 3 | 1 | 4 | 5 | 2 |
| 1 | 2 | 3 | 4 | 5 |

↑ Decryption

# Keyed Transposition Ciphers

- The third character in the plaintext block becomes the first character in the ciphertext block;
- The first character in the plaintext block becomes the second character in the ciphertext block; and so on.
- The permutation yields

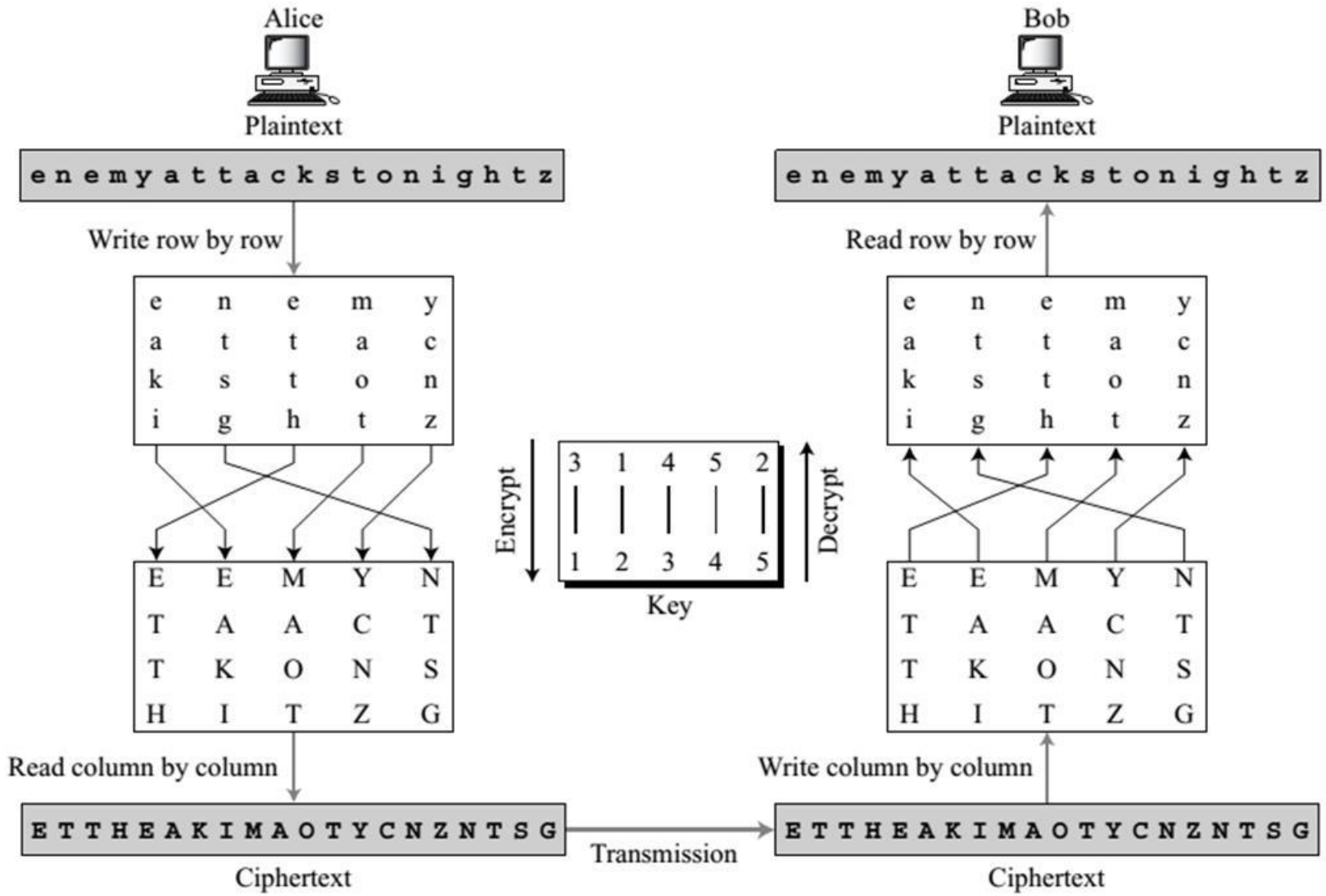
| | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **E** | **E** | **M** | **Y** | **N** | | **T** | **A** | **A** | **C** | **T** | | **T** | **K** | **O** | **N** | **S** | | **H** | **I** | **T** | **Z** | **G** |

- Alice sends the ciphertext “EEMYNTAACTTKONSHITZG” to Bob.
- Bob divides the ciphertext into 5-character groups and, using the key in the reverse order, finds the plaintext.

# Combining Two Approaches

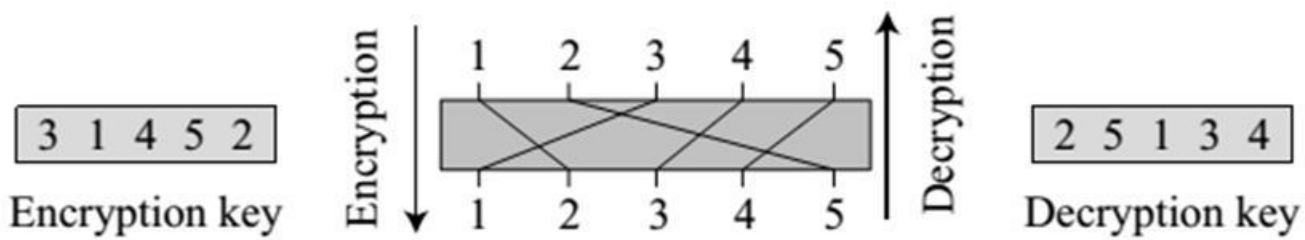
- More recent transposition ciphers combine the two approaches to achieve better scrambling.
- Encryption or decryption is done in three steps.
  - First, the text is written into a table row by row.
  - Second, the permutation is done by reordering the columns.
  - Third, the new table is read column by column.
- The first and third steps provide a keyless global reordering;
- The second step provides a blockwise keyed reordering.
- These types of ciphers are often referred to as keyed columnar transposition ciphers or just columnar transposition ciphers.

# Combining Two Approaches

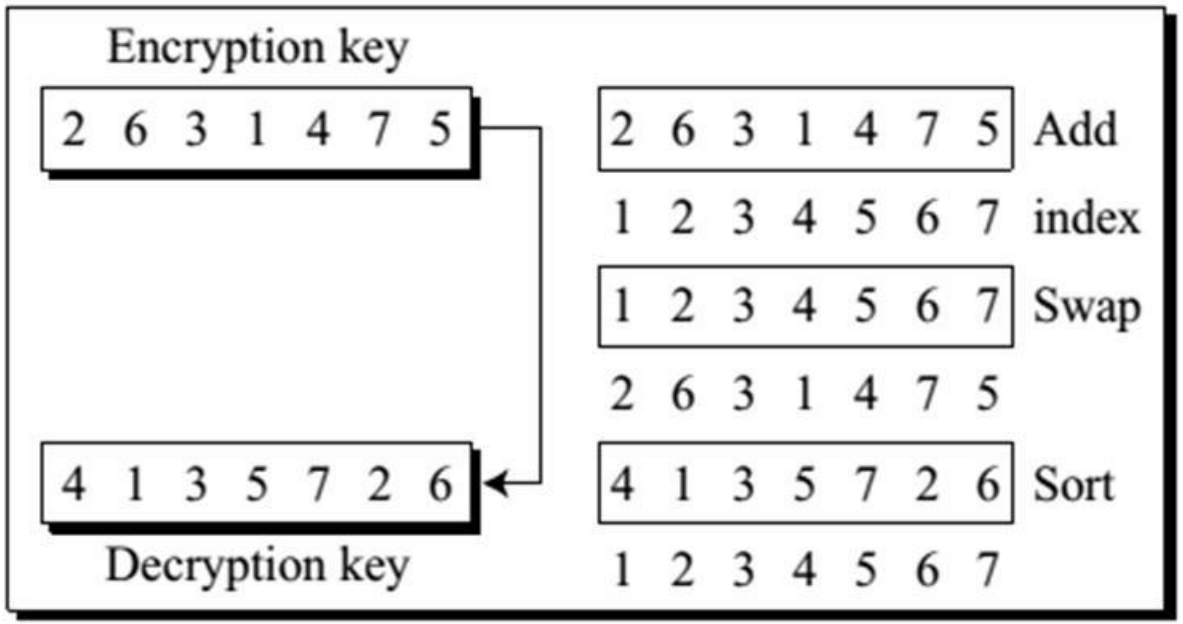


# Keys

- A single key was used in two directions for the column exchange: downward for encryption, upward for decryption.
- It is customary to create two keys from this graphical representation: one for encryption and one for direction.
- The keys are stored in tables with one entry for each column.
- The entry shows the source column number; the destination column number is understood from the position of the entry.
- The encryption key is (3 1 4 5 2). The first entry shows that column 3 (contents) in the source becomes column 1 (position or index of the entry) in the destination.
- The decryption key is (2 5 1 3 4). The first entry shows that column 2 in the source becomes column 1 in the destination.



# Key inversion in a transposition cipher




a. Manual process

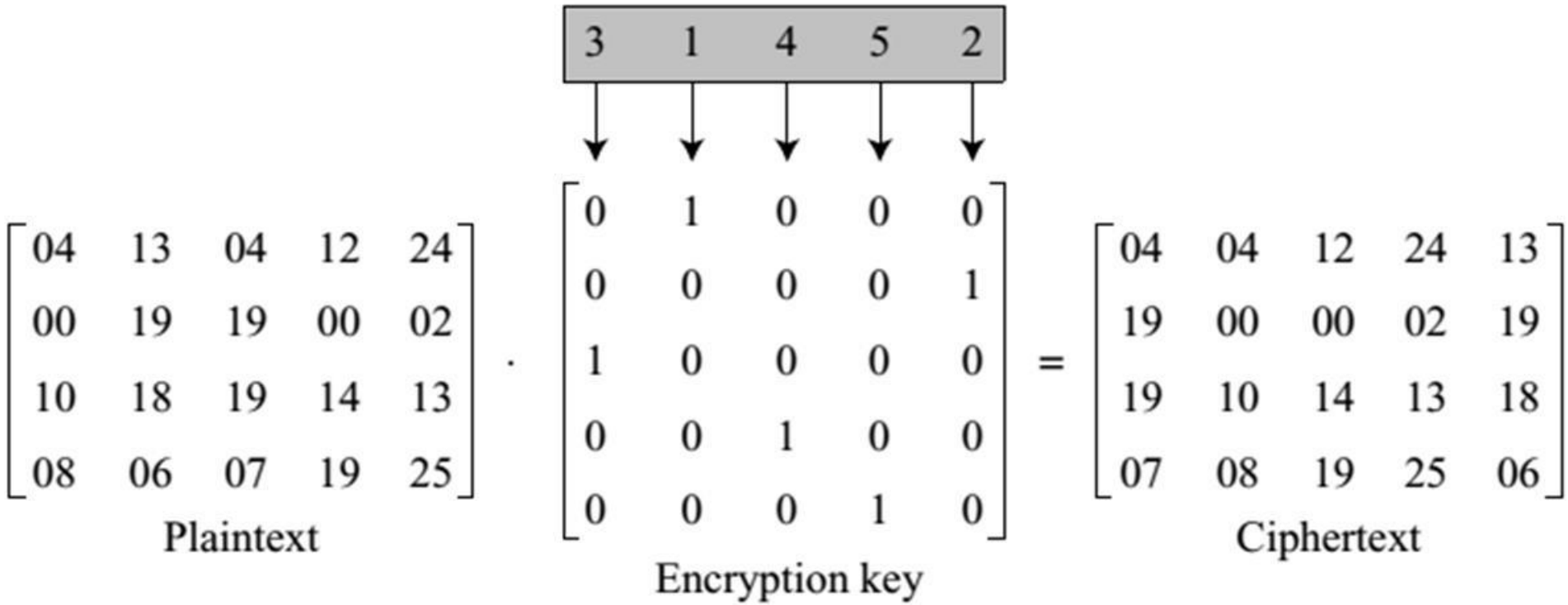
```
Given: EncKey [index]
index ← 1
while (index ≤ Column)
 {
  DecKey[EncKey[index]] ← index
  index ← index + 1
 }
Return : DecKey [index]
```

b. Algorithm

# Transposition Cipher: Using Matrices

- The plaintext and ciphertext are l × m matrices representing the numerical values of the characters;
- The keys are square matrices of size m × m.
- In a permutation matrix, every row or column has exactly one 1 and the rest of the values are 0s.
- Encryption is performed by multiplying the plaintext matrix by the key matrix to get the ciphertext matrix;
- Decryption is performed by multiplying the ciphertext by the inverse key matrix to get the plaintext matrix.
- A very interesting point is that the decryption matrix in this case is the inverse of the encryption matrix.
- However, there is no need to invert the matrix, the encryption key matrix can simply be transposed (swapping the rows and columns) to get the decryption key matrix.

# Representation of the key as a matrix in the transposition cipher




$$\begin{bmatrix} 04 & 13 & 04 & 12 & 24 \\ 00 & 19 & 19 & 00 & 02 \\ 10 & 18 & 19 & 14 & 13 \\ 08 & 06 & 07 & 19 & 25 \end{bmatrix} \cdot \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 04 & 04 & 12 & 24 & 13 \\ 19 & 00 & 00 & 02 & 19 \\ 19 & 10 & 14 & 13 & 18 \\ 07 & 08 & 19 & 25 & 06 \end{bmatrix}$$

# Cryptanalysis of Transposition Ciphers

- Transposition ciphers are vulnerable to several kinds of ciphertext-only attacks.
- Statistical Attack
  - First attack that can be applied is single-letter frequency analysis.
- Brute-Force Attack
  - Eve can try all possible keys to decrypt the message.
  - However, the number of keys can be huge (1! + 2! + 3! + . . . + L!), where L is the length of the ciphertext.
  - A better approach is to guess the number of columns.
  - Eve knows that the number of columns divides L. For example, if the length of the cipher is 20 characters, then $20 = 1 \times 2 \times 2 \times 5$.
  - This means the number of columns can be a combination of these factors (1, 2, 4, 5, 10, 20). However, the first (only one column) is out of the question and the last (only one row) is unlikely.

# Pattern Attack

- The ciphertext created from a keyed transposition cipher has some repeated patterns.

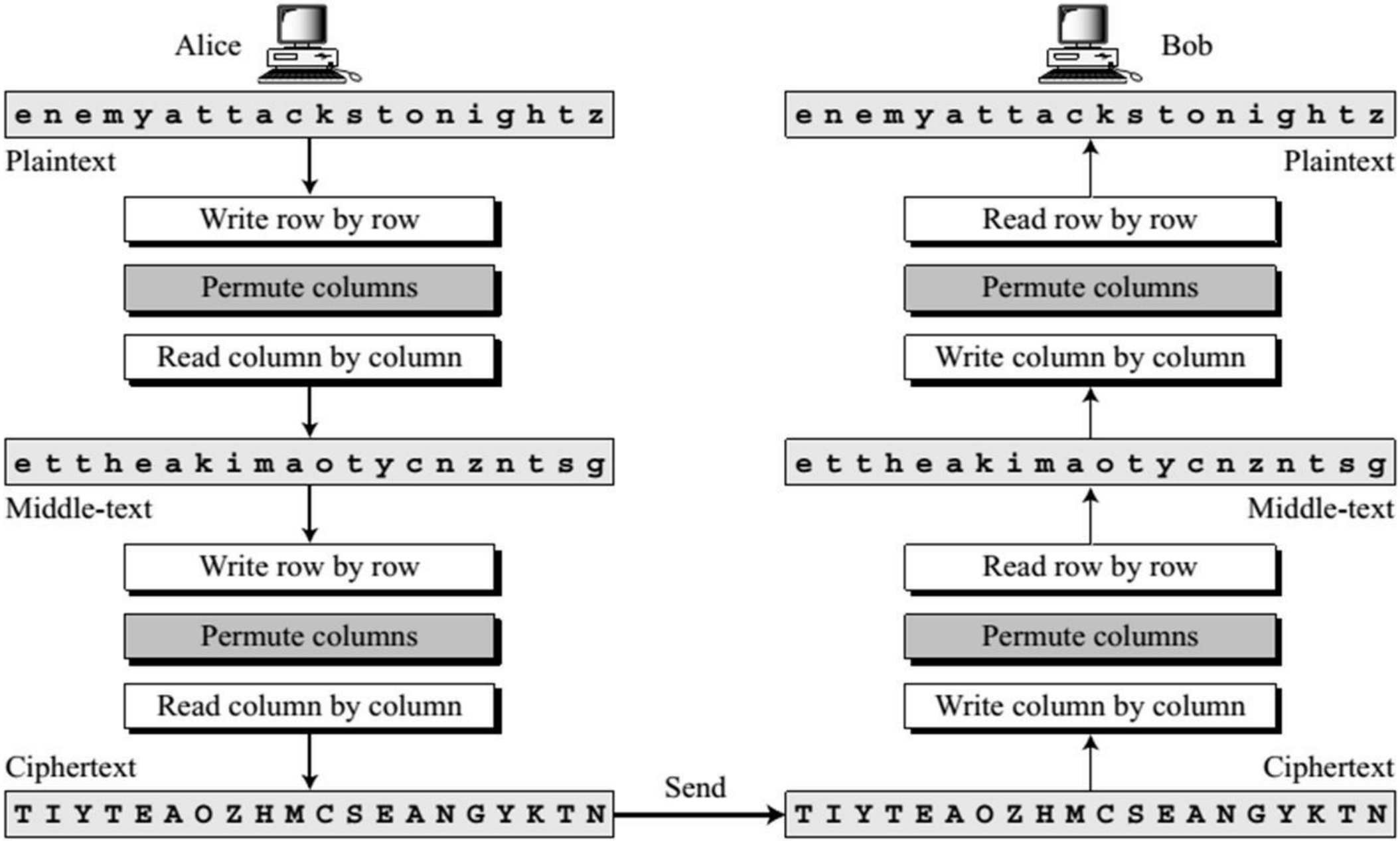
| | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **03** | **08** | **13** | **18** | *01* | *06* | *11* | *16* | **04** | **09** | **14** | **19** | *05* | *10* | *15* | *20* | **02** | **07** | **12** | **17** |

The 1st character in the ciphertext comes from the 3rd character in the plaintext.
The difference between the two adjacent numbers is 5.

This regularity can be used by the cryptanalyst to break the cipher.
If Eve knows or can guess the number of columns (which is 5 in this case), she can organize the ciphertext in groups of four characters.
Permuting the groups can provide the clue to finding the plaintext.

# Double Transposition Ciphers

- Double transposition ciphers can make the job of the cryptanalyst difficult.
- An example of such a cipher would be the one that repeats twice the algorithm used for encryption and decryption.
- A different key can be used in each step, but normally the same key is used.


Alice
e n e m y a t t a c k s t o n i g h t z
Plaintext
Write row by row
Permute columns
Read column by column
e t t h e a k i m a o t y c n z n t s g
Middle-text
Write row by row
Permute columns
Read column by column
Ciphertext
T I Y T E A O Z H M C S E A N G Y K T N
Send
Bob
e n e m y a t t a c k s t o n i g h t z
Plaintext
Read row by row
Permute columns
Write column by column
e t t h e a k i m a o t y c n z n t s g
Middle-text
Read row by row
Permute columns
Write column by column
Ciphertext
T I Y T E A O Z H M C S E A N G Y K T N

# Cryptanalysis of Transposition Ciphers

- Although, the cryptanalyst can still use the single-letter frequency attack on the ciphertext, a pattern attack is now much more difficult.
- The pattern analysis of the text shows

| | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 13 | 16 | 05 | 07 | 03 | 06 | 10 | 20 | 18 | 04 | 10 | 12 | 01 | 09 | 15 | 17 | 08 | 11 | 19 | 02 |

# STREAM AND BLOCK CIPHERS

- Symmetric ciphers into two broad categories: stream ciphers and block ciphers.
- Stream Ciphers
  - In a stream cipher, encryption and decryption are done one symbol (such as a character or a bit) at a time.
  - We have a plaintext stream, a ciphertext stream, and a key stream.
  - Call the plaintext stream P, the ciphertext stream C, and the key stream K.

$$P = P_1P_2P_3, \ldots \qquad C = C_1C_2C_3, \ldots \qquad K = (k_1, k_2, k_3, \ldots)$$

$$C_1 = E_{k1}(P_1) \qquad C_2 = E_{k2}(P_2) \qquad C_3 = E_{k3}(P_3) \ldots$$

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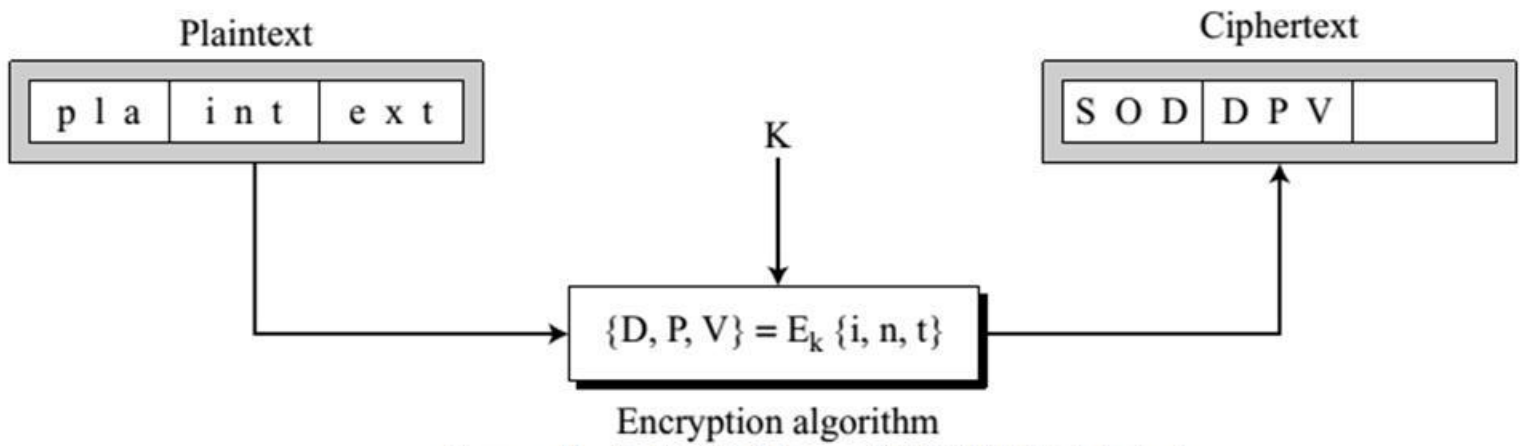
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# STREAM AND BLOCK CIPHERS

- Block Ciphers
  - In a block cipher, a group of plaintext symbols of size m (m > 1) are encrypted together creating a group of ciphertext of the same size.
  - Based on the definition, in a block cipher, a single key is used to encrypt the whole block even if the key is made of multiple values.
  - In a block cipher, a ciphertext block depends on the whole plaintext block.



# STREAM AND BLOCK CIPHERS

- Stream ciphers

- Additive ciphers
- Monoalphabetic substitution ciphers
- Vigenere ciphers

- Block ciphers

- Playfair ciphers
- Hill ciphers
- Polyalphabetic cipher